

The wedge product of the coordinate differentials dq_i, dp_i may be written

$$dq_i \wedge dp_i(\xi, \eta) = \xi_i \eta_{i+N_c} - \xi_{i+N_c} \eta_i = \xi^T J^{(i)} \eta,$$

where $J^{(i)}$ is the matrix which has zeros everywhere except for $(J^{(i)})_{i,i+N_c} = 1$, $(J^{(i)})_{i+N_c,i} = -1$. Summing these terms results in the *symplectic 2-form*, denoted ψ_S :

$$\psi_S = \sum_{i=1}^{N_c} dq_i \wedge dp_i(\xi, \eta) = \xi^T \left(\sum_{i=1}^{N_c} J^{(i)} \right) \eta = \xi^T J \eta.$$

Thus the matrix

$$J = \begin{bmatrix} \mathbf{0} & I \\ -I & \mathbf{0} \end{bmatrix}$$

defines a representation for the 2-form ψ_S . It is easy to verify that

$$\psi_S = \sum_{i < j} dz_i \wedge J_{ij} dz_j.$$

In general, a differential 2-form ψ is represented in coordinates by

$$\psi_z = \sum_{ij} a_{ij}(z) dz_i \wedge dz_j,$$

with matrix of coefficients $A(z) = (a_{ij}(z))$.

The *pull-back* of a differential 1-form ψ under a phase space mapping Φ is defined as the action of ψ after transformation by the Jacobian matrix of the mapping. It is written $\Phi^* \psi$, so

$$(\Phi^* \psi_z)(\xi) = \psi_{\Phi(z)}(\Phi'(z)\xi).$$

The pull-back of a differential 2-form $\psi_1 \wedge \psi_2$ is consequently defined as

$$\Phi^*(\psi_1 \wedge \psi_2) = (\Phi^* \psi_1) \wedge (\Phi^* \psi_2).$$

Given a differential 2-form ψ_z represented by the matrix $A(z) = (a_{ij}(z))$, the *pull-back* of ψ_z under Φ is defined by

$$\Phi^* \psi_z = \sum_{ij} b_{ij}(z) dz_i \wedge dz_j,$$